

BPL_TEST2_Chemostat - demo

```
In [1]: # Import the application model and context variables
        from BPL_TEST2_Chemostat_fmpy import*
```

Windows - run FMU pre-compiled JModelica 2.14

```
In [2]: # Import the general FMU_explore functions and adapt them to the application
        from fmu_explore_fmpy import AdaptWith

        Adapted = AdaptWith(parValue, parLocation, parCheck, keyVariables, \
                               model_description, fmu_model, fmu_process_diagram, \
                               MSL_usage, MSL_version, BPL_version, \
                               opts_std, simulationTime, timeDiscreteStates, \
                               diagrams, ax, lines, external_function = cstrProdMax)

        par = Adapted.par
        init = Adapted.init
        disp = Adapted.disp
        simu = Adapted.simu
        show = Adapted.show

        resetPen = Adapted.resetPen
        process_diagram = Adapted.process_diagram
        describe_general = Adapted.describe_general
        describe_parts = Adapted.describe_parts
        describe_MSL = Adapted.describe_MSL
        cstrProdMax = Adapted.external_function

        FMU_explore_info = Adapted.FMU_explore_info
        system_info = Adapted.system_info
        SDG = Adapted.SDG
```

```
In [3]: run -i BPL_TEST2_Chemostat_fmpy_explore.py
```

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation and the command process_diagram() brings up the main configuration

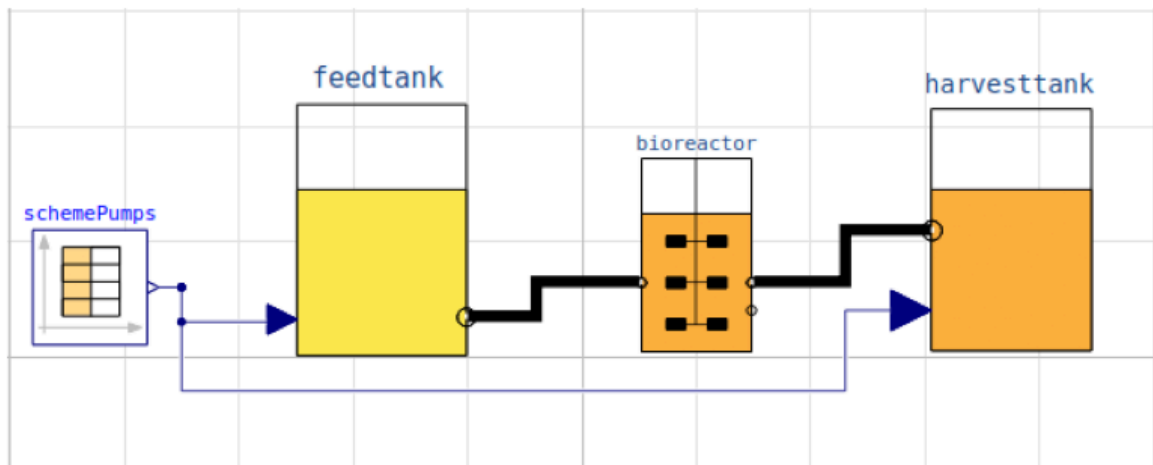
Brief information about a command by help(), eg help(simu)

Key system information is listed with the command system_info()

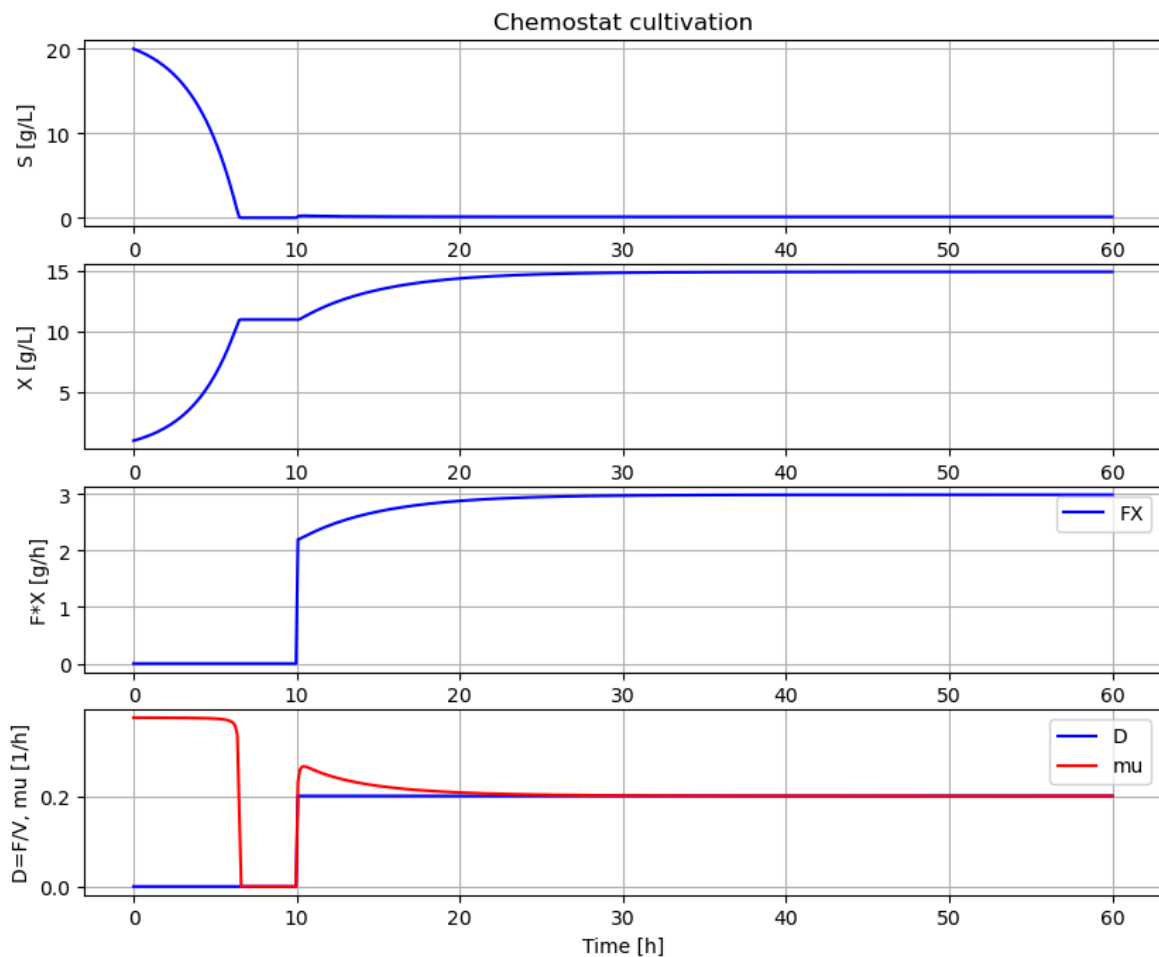
```
In [4]: %matplotlib inline
        plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

```
In [5]: process_diagram()
```

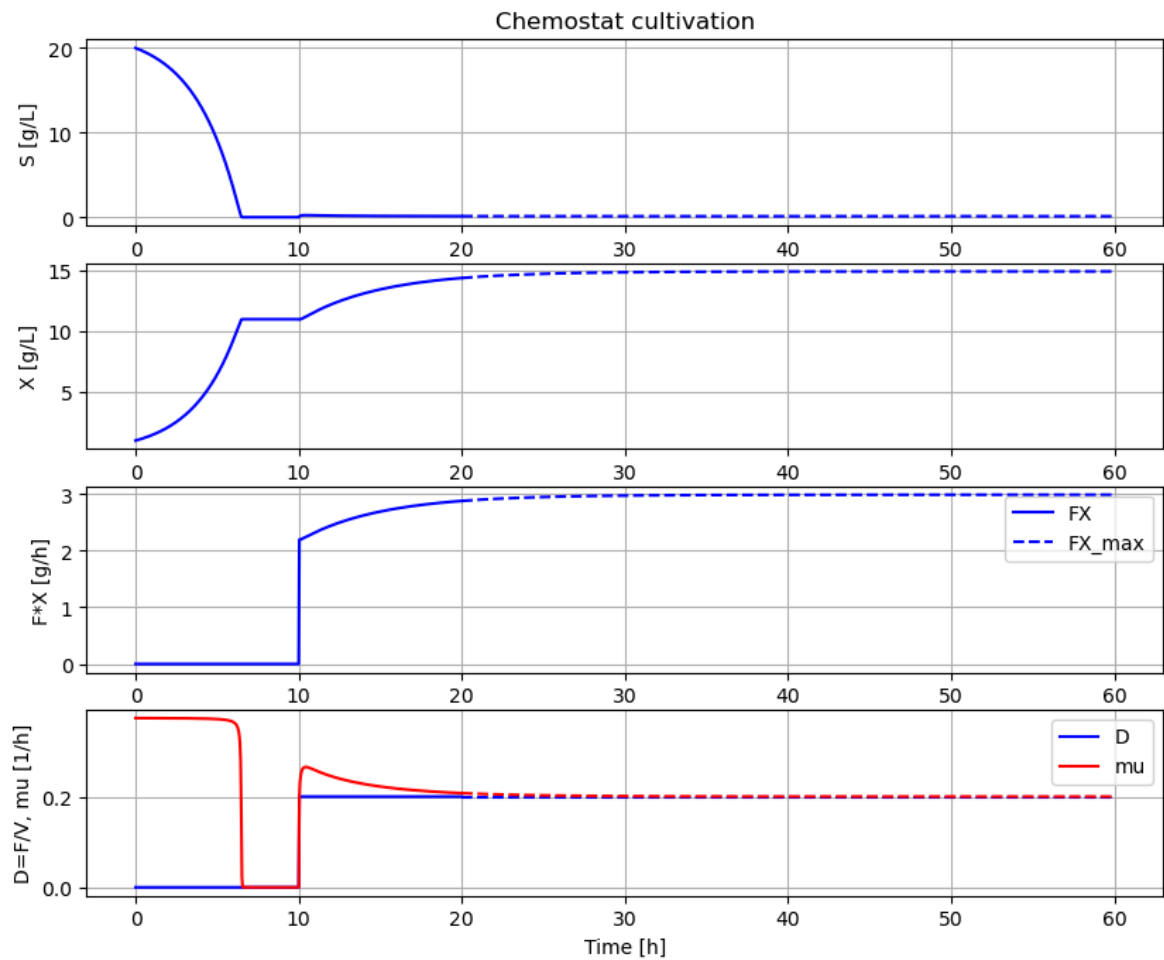
No processDiagram.png file in the FMU, but try the file on disk.



```
In [6]: newplot()
        par(Y=0.50, qSmax=0.75, Ks=0.1)           # Culture parameters
        init(V_start=1.0, VX_start=1.0, VS_start=20) # Bioreactor startup
        par(S_in=30, t0=0, F0=0, t1=10, F1=0.2)    # Substrate feeding
        simu(60)
```



```
In [7]: # Check simu('cont')
        newplot()
        simu(20);
        simu(40, 'cont')
```



```
In [8]: disp('culture')
```

```
Y : 0.5
qSmax : 1.0
Ks : 0.1
```

```
In [9]: describe('mu')
```

```
Cell specific growth rate variable : 0.2 [ 1/h ]
```

```
In [10]: describe('parts')
```

```
['bioreactor', 'bioreactor.culture', 'D', 'feedtank', 'harvesttank', 'MSL', 'schemePumps']
```

```
In [11]: describe('MSL')
```

```
MSL: RealInput, RealOutput, CombiTimeTable, Types
```

```
In [12]: system_info()
```

System information

- OS: Windows
- Python: 3.12.11
- Scipy: 1.16.0
- FMPy: 0.3.27
- FMU by: JModelica.org
- FMI: 2.0
- Type: CS
- Name: BPL.Examples_TEST2.Chemostat
- Generated: 2025-07-26T09:39:44
- MSL: 3.2.2 build 3
- Description: Bioprocess Library version 2.3.1
- Interaction: FMU_explore version 1.1.8